Winginx troubleshooting guide

502 errors, nginx conflicts, and PHP updates (Windows Server / winginx)

Document version: 2026-10-26 (rev 2)
Install root (default): `C:\winginx`
Repository: `C:\winginx\winginx-installer`
Recommended branch: `fix/install-in-place-upgrade`

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0. One-command audit (recommended first)

Elevated PowerShell — updates PHP via the patch script, **does not modify** `conf.d`, restarts services, and writes a log file for support:

```
cd C:\winginx\winginx-installer
git fetch origin
git checkout fix/install-in-place-upgrade
git pull

powershell -NoProfile -ExecutionPolicy Bypass -File .\scripts\troubleshoot-
winginx.ps1 `
  -InstallRoot C:\winginx
```

Log file (default): `C:\winginx\logs\winginx-troubleshoot-YYYYMMDD-HHmmss.log`

Optional:

```
# Custom log path
powershell -NoProfile -ExecutionPolicy Bypass -File .\scripts\troubleshoot-
winginx.ps1 `
  -InstallRoot C:\winginx `
  -LogPath C:\winginx\logs\support-audit.log

# Diagnostics only (no patch)
powershell -NoProfile -ExecutionPolicy Bypass -File .\scripts\troubleshoot-
winginx.ps1 `
  -InstallRoot C:\winginx -SkipPatch
```

Each section in the log includes a **COMMAND:** line you can copy and run manually.

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1. What HTTP 502 usually means

A **502 Bad Gateway** from nginx almost always means:

- nginx is running and accepted the request, but
- the **PHP FastCGI backend** (`php-cgi` on `127.0.0.1:9000`) did not respond correctly, or
- the **wrong virtual host** handled the request (wrong root / no PHP).

It is **not** the same as php2shell (CVE-2026-109618 / CVE-2026-109619). You can be on PHP 8.2.35 and still see 502 if nginx or vhosts are misconfigured.

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2. Quick health check (elevated PowerShell)

Run:

```
Get-Service winginx-nginx, winginx-php
Get-Process nginx, php-cgi -ErrorAction SilentlyContinue | Format-Table Id,
ProcessName -AutoSize
Test-NetConnection 127.0.0.1 -Port 9000
Test-Path C:\winginx\php\php.ini
Get-Content C:\winginx\php\logs\php-service.err.log -Tail 30
Get-Content C:\winginx\php\logs\php-service.out.log -Tail 30
```

How to read the results

Check	Good	Problem
Services	Both **Running**	Stop/start helpers (Section 4)
`nginx` process count	Small (about 2–9)	**Many nginx.exe** → multiple masters (Section 4)
`php-cgi`	At least **1**	PHP service down → 502 on `.php`
Port **9000**	`TcpTestSucceeded : True`	PHP not listening → 502
`php.ini`	**True**	Run Section 5 to create winginx `php.ini`

Note: Searching `error.log` for `502` can match **process IDs** (e.g. `50256#10856`), not HTTP 502. Use Section 6 for better log searches.

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3. Common log messages

3.1 `conflicting server name "…" on 0.0.0.0:8081, ignored`

Several files in `C:\winginx\nginx\conf\conf.d\` define the **same** `listen` port and **same** `server_name`. Nginx keeps the **first** block and **ignores** the rest.

Impact: Sites like `liptong.pdic.gov.ph` may not use the config you expect → wrong document root, missing PHP, or 502.

Fix: Section 5.

3.2 `CreateFile() "C:/winginx/nginx/logs/nginx.pid" failed`

Something ran `nginx -s reload` or `nginx -s quit` while **no master** was running with prefix `C:\winginx\nginx` (crashed, never started, or wrong start command).

Fix: Section 4 (clean stop/start). After a good start, `C:\winginx\nginx\logs\nginx.pid` should exist.

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4. Stabilize nginx (fix multiple `nginx.exe` processes)

Run in **elevated** PowerShell:

```
powershell -ExecutionPolicy Bypass -File C:\winginx\winginx-installer\scripts\stop-
nginx.ps1
Start-Sleep -Seconds 3
Get-Process nginx -ErrorAction SilentlyContinue | Stop-Process -Force
Start-Sleep -Seconds 2
(Get-Process nginx -ErrorAction SilentlyContinue).Count
```

The last command should print **0**.

Ensure the logs directory exists, then start:

```
New-Item -ItemType Directory -Force C:\winginx\nginx\logs | Out-Null
powershell -ExecutionPolicy Bypass -File C:\winginx\tools\start-nginx.ps1
Start-Sleep -Seconds 2
(Get-Process nginx).Count
Test-Path C:\winginx\nginx\logs\nginx.pid
```

Expect: A **small** nginx process count (not 15+) and `nginx.pid` **True**.

Use **stop-nginx.ps1** / **start-nginx.ps1** for restarts. Avoid ad-hoc `nginx -s reload` until `nginx.pid` exists.

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5. Fix duplicate virtual hosts on port 8081

List configs that mention port 8081 or your site name:

```
Get-ChildItem C:\winginx\nginx\conf\conf.d\*.conf | ForEach-Object {
  $hits = Select-String -Path $_.FullName -Pattern
'8081|liptong\.pdic\.gov\.ph|server_name'
  if ($hits) { "`n=== $($_.Name) ==="; $hits.Line }
}
```

Action:

1. For each `listen 8081` + `server_name` combination, keep **one** `.conf` file.
2. Disable duplicates by renaming (example: `site.conf` → `site.conf.disabled`).
3. Test configuration:

```
cd C:\winginx\nginx
.\nginx.exe -p C:\winginx\nginx -c conf\nginx.conf -t
```

4. Restart with **start-nginx.ps1** (Section 4).

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6. Find real 502 and upstream errors in logs

```
Select-String -Path C:\winginx\nginx\logs\error.log `
  -Pattern 'upstream|connect\(\) failed|recv\(\) failed|Primary script unknown' |
  Select-Object -Last 20
```

```
Select-String -Path C:\winginx\nginx\logs\access.log -Pattern ' 502 ' |
  Select-Object -Last 20
```

Note the **time**, **Host**, and **URL** when users report 502.

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7. Test the site on the server (bypass front proxy)

```
Invoke-WebRequest -Uri 'http://127.0.0.1:8081/' `
  -Headers @{ Host='liptong.pdic.gov.ph' } -UseBasicParsing
```

Replace host and URL as needed.

Result	Meaning
502 on server	Fix winginx (Sections 4–5, PHP logs)
OK on server, 502 in browser	Check load balancer, IIS, firewall, or DNS in front of winginx

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8. PHP / php2shell updates

Important: Do **not** re-run old `install.ps1` builds that delete all `conf.d\*.conf`. Use branch `fix/install-in-place-upgrade`, the patch script, or `install.ps1` with **`-SkipNginxReplace`** (preserves custom vhosts by default on current branch).

8.1 PHP-only security patch (recommended on live servers)

Does **not** replace nginx binaries or rewrite full `php.ini` (keeps existing file when present):

```
powershell -ExecutionPolicy Bypass -File C:\winginx\winginx-installer\scripts\stop-
nginx.ps1
powershell -ExecutionPolicy Bypass -File C:\winginx\winginx-installer\scripts\patch-
php-php2shell.ps1 `
  -InstallRoot C:\winginx
powershell -ExecutionPolicy Bypass -File C:\winginx\tools\start-nginx.ps1
```

If output says **PHP 8.2.35** but **`php.ini` was not found**, run Section 8.2.

8.2 Create or refresh `php.ini` without replacing nginx

Use branch `fix/install-in-place-upgrade` (or current main with `-SkipNginxReplace`):

```
cd C:\winginx\winginx-installer
git fetch origin
git checkout fix/install-in-place-upgrade
git pull

powershell -ExecutionPolicy Bypass -File .\scripts\stop-nginx.ps1
```

```
powershell -NoProfile -ExecutionPolicy Bypass -File .\install.ps1 `
  -InstallRoot C:\winginx `
  -PhpVersion 8.2.35 `
  -SkipNginxReplace

powershell -ExecutionPolicy Bypass -File C:\winginx\tools\start-nginx.ps1
```

Verify:

```
Test-Path C:\winginx\php\php.ini
Select-String -Path C:\winginx\php\php.ini -Pattern 'session.upload_progress'
C:\winginx\php\php-cgi.exe -v
```

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9. If `install.ps1` fails on `C:\winginx\nginx` (Access denied)

Stop services first. Prefer `-SkipNginxReplace` (Section 8.2) or the **patch script** (Section 8.1) instead of replacing the whole nginx folder.

Remove leftover stale folders only when services are stopped:

```
Remove-Item C:\winginx\nginx.stale-* -Recurse -Force -ErrorAction SilentlyContinue
```

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10. Suggested order of operations

1. Section 2 — quick health check
2. Section 4 — clean nginx process sprawl
3. Section 5 — dedupe vhosts on 8081
4. Section 6–7 — confirm 502 on box vs in browser
5. Section 8 — PHP 8.2.35 + `php.ini` if needed
6. Re-test the affected site

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11. Support information to collect

When escalating, provide:

- Output of Section 2 (service status, process counts, port 9000, `php.ini`)
- `(Get-Process nginx).Count` after Section 4
- Output of `nginx -t`
- Last 20 lines from Section 6 (access + error log)
- Host header test result (Section 7)

Generated for winginx-installer operators on Windows Server.